NYC YELLOW TAXI ANALYTICS PROJECT

Business Requirement Document (BRD) + Interview Guide

Prepared for: Data Analyst Portfolio and Interview Readiness

Project Scope: Revenue proxy, demand, zone, payment, vendor, airport, and time-based operations analytics

Dataset Type: Real public anonymized trip records from the NYC TLC Yellow Taxi program

Target Role: 1 year to 3 year Data Analyst profile

Document Version	1.0
Business Use Case	Executive reporting, operational improvement, demand planning, and service quality analysis
Primary Outputs	BRD, FRD, KPI definitions, star schema, SQL tasks, Python analysis, and dashboard requirements
Project Positioning	Consulting-style enterprise analytics case study using a real public government dataset

This document frames the public NYC Taxi dataset as a real enterprise analytics engagement. The scenario, requirements, KPIs, and interview questions are written in consulting style so you can present the project like a client assignment rather than a casual practice exercise.

1. Executive Summary

The NYC Yellow Taxi business operates in a highly dynamic mobility environment where demand, trip duration, zone activity, payment behavior, airport traffic, and vendor utilization change by hour, day, and season. Leadership needs a single analytics layer to understand what is happening across the network, why it is happening, and where operational attention should be directed.

This project is designed as an enterprise BI case study using the official NYC TLC Yellow Taxi trip records. The objective is to convert monthly public trip data into a decision-support model that helps leadership monitor revenue proxy, service demand, airport contribution, payment mix, tip behavior, and vendor-level performance in one place.

The final deliverable should support executive reporting, operations management, and analytical exploration in Excel, SQL, Python, and Power BI. The analysis should be able to answer both broad management questions and detailed diagnostic questions without claiming access to proprietary company systems or confidential business metrics.

2. Business Background and Current Situation

- Trip volume is highly concentrated in certain hours, boroughs, and zones, but the demand drivers are not visible in a single report.
- Airport trips, long-distance trips, and high-tip trips may contribute disproportionately to revenue proxy, yet this is not consistently monitored.
- Payment behavior is fragmented across cash, card, and other payment modes, creating limited visibility into customer preferences.
- Vendor productivity and zone productivity are often reviewed separately, which makes it difficult to compare operational efficiency across the network.
- Seasonal patterns, day-of-week patterns, and night vs. day demand shifts are not standardized in management reporting.
- Exception trips, such as zero-distance records, suspicious durations, or outlier fare values, need to be identified and filtered before executive reporting.

In the current situation, the business has enough raw trip data to make informed decisions, but it lacks a structured analytics model that converts the raw data into KPI-driven insight for executives and operations teams.

3. Problem Statement

The organization requires a centralized analytics solution to analyze historical taxi trip, payment, zone, vendor, airport, and time-based data. The solution must identify the business drivers of demand, revenue proxy, trip efficiency, payment behavior, and service patterns, while also highlighting operational bottlenecks and exception records. The final output should provide leadership with clear KPIs and actionable insights that can improve fleet utilization, zone planning, and service quality.

4. Business Objectives

- Measure total trips, total fare value, and overall revenue proxy across the selected analysis window.
- Understand demand by hour, day of week, month, and season to support fleet planning.
- Evaluate zone-level and borough-level trip concentration to identify high-traffic hotspots.
- Measure airport trip contribution and understand whether airport travel is an important revenue driver.
- Analyze trip distance, trip duration, and fare efficiency to identify operational patterns.
- Track payment type distribution and cash vs. card mix to understand transaction behavior.
- Measure tip behavior and identify conditions associated with higher or lower tip rates.

- Compare vendor performance using trip volume, revenue proxy, and trip characteristics.
- Identify exception records and outliers that should be excluded or separately tracked in management reporting.
- Build a Power BI dashboard that supports executive decision making and operations monitoring.

5. Project Scope

In Scope	Trip analysis, revenue proxy analysis, payment analysis, zone analysis, airport trips, vendor analysis, time intelligence, exception filtering, Power BI dashboard, SQL analysis, Python EDA
Out of Scope	Driver payroll, dispatch optimization, route optimization, forecasting model development, accounting-grade profitability, fraud analytics, and confidential internal company data
Future Scope	Forecasting, weather integration, congestion analysis, driver productivity scoring, and interactive geospatial modeling

6. Stakeholders and Expected Usage

Stakeholder	Primary Need	How They Use the Output
CEO / Executive Team	Strategic performance review	Monitors revenue proxy, growth patterns, airport contribution, and operational health
Operations Lead	Demand and service control	Reviews peak hours, zone demand, delays, and exception trends
Revenue / Finance Lead	Trip value analysis	Tracks fare mix, tip behavior, and payment mode contribution
Fleet Manager	Utilization and coverage	Uses demand by time and zone to improve allocation
BI / Data Analyst	Reporting and insight generation	Builds dashboards, SQL outputs, and recurring reports

7. Data Source and Trustworthiness

The project uses the public NYC TLC Yellow Taxi trip record data, which is published as an official government dataset and accompanied by a data dictionary and taxi zone lookup resources. Because the data is public, anonymized, and widely used for analytics and research, it is appropriate for portfolio work, training, and interview demonstrations.

The project should clearly state that the business scenario is fictional while the data is real. The analysis must not claim proprietary access to company systems, customer identities, driver identities, or internal financial statements.

8. Assumptions and Constraints

- Total amount or fare amount can be used as a revenue proxy, but it is not the same as accounting revenue unless explicitly defined that way.
- Profit is not directly available and should not be reported as a native metric unless additional cost assumptions are introduced.
- Customer-level identity is not available in the taxi data, so analysis should focus on trips, zones, vendors, time, and payment behavior rather than customer retention.
- The data is historical and best suited for descriptive and diagnostic analysis.
- 2025 and later records include a congestion fee column, so the model should handle version-specific fields carefully.
- Outliers such as extremely short trips, zero values, or abnormal durations should be filtered or flagged during analysis.

9. Data Fields Used in the Analysis

Field	Business Use
VendorID	Vendor comparison and utilization analysis
tpep_pickup_datetime	Pickup timing, hour-of-day trend, day-of-week trend
tpep_dropoff_datetime	Trip duration and operational efficiency
passenger_count	Ride characteristics and exception checks
trip_distance	Distance-based operational analysis
RatecodeID	Rate type, airport trips, special trip categories
PULocationID / DOLocationID	Pickup and dropoff zone intelligence
payment_type	Cash vs card mix and payment behavior
fare_amount, tip_amount, total_amount	Revenue proxy, tip behavior, and trip value analysis

10. Functional Requirements (FRD Summary)

Module	Requirement
Revenue Analysis	Show total trips, fare proxy, total amount, average fare, and monthly trend.
Demand Analysis	Show trip volume by hour, day, week, month, and season.
Zone Analysis	Show top pickup and dropoff zones, borough contribution, and zone concentration.
Airport Analysis	Track airport trips, share of total trips, and revenue contribution by airport-related rate codes.
Payment Analysis	Show payment type distribution, cash vs card mix, and tip differences by payment mode.
Vendor Analysis	Compare vendors by trip count, value, and average trip characteristics.
Exception Analysis	Flag suspicious, zero-value, or outlier trip records for data quality review.

11. Data Model and Star Schema

Recommended model: one central fact table with supporting dimensions. This structure simplifies BI reporting, reduces join complexity, and supports fast analytical queries.

Table Type	Suggested Name	Purpose
Fact Table	fact_yellow_trip	One row per taxi trip with time, zone, vendor, payment, fare, distance, and duration context
Dim Date	dim_date	Calendar attributes for daily, monthly, quarterly, and yearly reporting
Dim Time	dim_time	Hour-of-day and time-band analysis
Dim Zone	dim_zone	Taxi zone, borough, and service area analysis
Dim Payment	dim_payment_type	Cash, card, and other payment mode analysis
Dim Rate Code	dim_rate_code	Airport, standard, and special rate analysis
Dim Vendor	dim_vendor	Vendor-level comparison and utilization analysis

Primary keys should be assigned to the dimension tables, while the fact table should carry foreign keys back to the date, time, zone, payment, rate code, and vendor dimensions. The fact table should remain granular at the trip level so that all executive metrics can be rolled up without losing detail.

12. KPI Definitions

KPI	Business Meaning
Total Trips	Total number of taxi trips in the analysis window
Total Revenue Proxy	Sum of total_amount or fare-related value, clearly labeled as proxy
Average Fare Amount	Average fare per trip
Average Tip Amount	Average tip per trip
Average Trip Distance	Average trip distance in miles
Average Trip Duration	Average minutes between pickup and dropoff

KPI	Business Meaning
Revenue per Mile	Revenue proxy divided by trip distance
Tip Rate	Tip amount as a share of trip value
Airport Trip Share	Share of trips associated with airport rate codes
Cash vs Card Mix	Payment distribution by payment type
Peak Hour Trips	Trip count by hour of day
Vendor Share	Share of trips contributed by each vendor

13. Core Business Questions

1. Which hours of the day generate the highest trip volume?
2. Which days of the week have the strongest demand patterns?
3. Which months show the highest and lowest trip counts?
4. Which boroughs and zones contribute the most trips?
5. Which pickup zones and dropoff zones are the busiest?
6. Which airport-related trips contribute the most to total trip value?
7. How does trip distance vary by zone and time band?
8. How does trip duration differ between day and night operations?
9. Which payment type is most common?
10. Are card-paying trips associated with higher tips than cash-paying trips?
11. Which vendors handle the highest number of trips?
12. Which vendors have the highest average fare or trip value?
13. Which zones show the highest revenue proxy per mile?
14. Which trips are likely to be exceptions or outliers?
15. How do weekday patterns differ from weekend patterns?
16. What is the relationship between trip distance and tip amount?
17. Which time bands should receive more operational attention?
18. Which boroughs drive the strongest demand by hour and by month?
19. Are airport trips more valuable than regular street-hail trips?
20. What operational patterns are associated with high-value trips?

14. Success Criteria

- A clean star schema is built and validated.
- All KPI definitions are documented and consistently calculated across Excel, SQL, Python, and Power BI.
- Executive dashboards answer the main business questions without manual intervention.
- Data quality rules are applied to flag or remove suspicious trip records.
- Insights are actionable for operations, revenue, vendor management, and fleet planning.
- The project can be clearly explained in an interview as a real-data analytics case study.

15. Power BI Dashboard Requirement Specification

Page	Required Content
Page 1 - Executive Overview	Headline KPIs, monthly trend, revenue proxy, trip count, and operational health
Page 2 - Demand Analysis	Hour-of-day trend, day-of-week trend, monthly seasonality, and peak time bands
Page 3 - Zone Analysis	Top pickup and dropoff zones, borough comparison, and zone concentration
Page 4 - Revenue and Fare Analysis	Fare amount, total amount, distance efficiency, and high-value trip patterns

Page	Required Content
Page 5 - Payment and Tips	Payment mix, cash vs card share, tip amount, and tip rate comparisons
Page 6 - Vendor and Airport	Vendor ranking, airport trip share, rate code analysis, and service pattern monitoring
Page 7 - Exception Monitoring	Outlier trips, zero-value records, suspicious duration checks, and data quality flags

16. SQL Task Book - Suggested Practice Areas

- Write joins across fact, date, time, zone, payment, rate code, and vendor tables.
- Calculate monthly trip trends using GROUP BY and date functions.
- Find top 10 pickup zones by trip count and revenue proxy.
- Measure airport trip share by month and by vendor.
- Find vendors with the highest and lowest average trip value.
- Analyze peak-hour demand using window functions and ranking.
- Create trip duration buckets and compare them across zones.
- Compare weekday and weekend trip patterns.
- Build exception filters for suspicious or invalid trip records.

17. Python Analysis Book - Suggested Practice Areas

- Load the monthly parquet files and the taxi zone lookup data.
- Check missing values, duplicates, and data types.
- Create derived fields such as trip duration, hour band, weekday, and month.
- Perform exploratory analysis by zone, borough, vendor, payment type, and time band.
- Visualize trends using line charts, bar charts, histograms, and heatmaps.
- Build summary tables for Power BI and SQL validation.
- Flag outliers such as zero-distance trips, abnormal durations, and unusual fare values.

18. Interview Q&A; Guide

Q. What is the business goal of this project?

A. To understand trip demand, revenue proxy, zone activity, payment behavior, vendor performance, and operational patterns in one integrated view so management can improve fleet utilization and service quality.

Q. Why is NYC Taxi a strong dataset for analytics?

A. It is a real public government dataset with high volume, clear relational structure, and enough business context to support SQL, Python, Excel, and Power BI work.

Q. What is the main problem statement?

A. The company needs a centralized analytics layer to identify demand drivers, operational bottlenecks, payment behavior, and zone-level performance.

Q. Why did you choose a star schema?

A. A star schema simplifies reporting, improves query performance, and makes dashboard development easier for analytical use cases.

Q. What are the key KPIs?

A. Total trips, revenue proxy, average fare, average trip distance, average trip duration, tip rate, airport trip share, and payment mix.

Q. What is the most important business insight you expect?

A. Peak-hour demand and zone concentration strongly influence operational planning, while airport trips and payment mode provide additional value context.

Q. What is the limitation of the dataset?

A. It does not contain customer identity, driver identity, or accounting-grade profit data, so the analysis should focus on trip-level operations and revenue proxy.

Q. How would you explain revenue?

A. In this project, total amount or fare-related totals can be used as a revenue proxy, but that definition must be stated clearly in the documentation.

Q. How would the dashboard be used by the CEO?

A. To monitor growth, demand peaks, airport contribution, vendor comparison, and service quality trends.

Q. Why is this project interview-worthy?

A. It shows end-to-end thinking: business framing, data modeling, data quality checks, KPI design, SQL analysis, Python EDA, and dashboard storytelling.

19. Resume-Ready Summary

Built an enterprise-style NYC Yellow Taxi analytics case study using the public NYC TLC trip record dataset. Defined a business requirement document, proposed a star schema, documented KPIs, and prepared reporting requirements for SQL, Python, and Power BI.

20. Appendix - How to Present This Project in an Interview

- Start with the business problem, not the dataset.
- Explain that the scenario is fictional but the data is real and public.
- Describe the fact table and the supporting dimension tables clearly.
- Mention 3 to 5 KPIs and define them in business language.
- Call out data limitations honestly, especially the absence of customer identity and profit data.
- End with the operational and revenue proxy insights you would expect to deliver.

This project can be used as a structured portfolio case study for interviews, LinkedIn, and resume bullets.